

Early Use of Blood Flow Restriction Training with Low-Intensity Exercises following Anterior Cruciate Ligament Reconstruction Improves Quadriceps Strength and Post-Operative Pain: A Randomized Controlled Trial

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Objectives:

Blood flow restriction therapy (BFRT) has been proposed as a method to enhance rehabilitation following anterior cruciate ligament reconstruction (ACLR). However, a paucity of data exists to support the use of BFRT in clinical practice. The purpose of this study is to determine if early application of BFRT, used with low-weight therapy exercises, increases quadriceps strength, and functional outcomes following ACLR compared to a traditional rehabilitation protocol without BFRT.

Methods:

Twenty-six patients undergoing ACLR were randomized to receive either: (1) a traditional rehabilitation program (n=14), or (2) a modified program using BFRT with low-weight (20-50% of 1-repetition maximum) exercises (n=12). Two patients crossed over from the control to BFRT groups. An as treated analysis was performed for biweekly measurements in the acute postoperative period of range of motion (ROM), thigh circumference, and terminal knee extension (TKE) strength. Circumference and TKE strength were analyzed as a percentage of the contralateral side. VAS and IKDC scores were assessed preoperatively and during the first 12 weeks postoperatively.

Results:

Compared to the control group, the BFRT patients demonstrated significantly greater TKE strength at week 8 (68.3% vs. 82.7%, $p = 0.022$), week 10 (69.0% vs. 82.8%, $p = 0.044$), and week 12 (64.4% vs. 84.2%, $p = 0.002$), as well as greater overall change in TKE strength from week 3 to 12 (8.3% vs 24.4%, $p = 0.044$). The BFRT group reported significantly lower VAS values at week 6 (2.5 vs. 0.6, $p = 0.041$), though scores were similar at week 12 (0.9 vs. 1.0, $p = 0.912$). There was no significant difference for knee ROM, thigh circumference, or IKDC scores at any time point.

Conclusions:

In comparison to a traditional rehabilitation protocol following ACLR, BFRT used in conjunction with low-weight exercises during the acute postoperative period was associated with significantly lower VAS scores at week 6, as well as greater quadriceps strength at 8, 10, and 12 weeks. These results suggest that BFRT may accelerate rehabilitation after ACLR, though future studies are needed to assess long-term outcomes.

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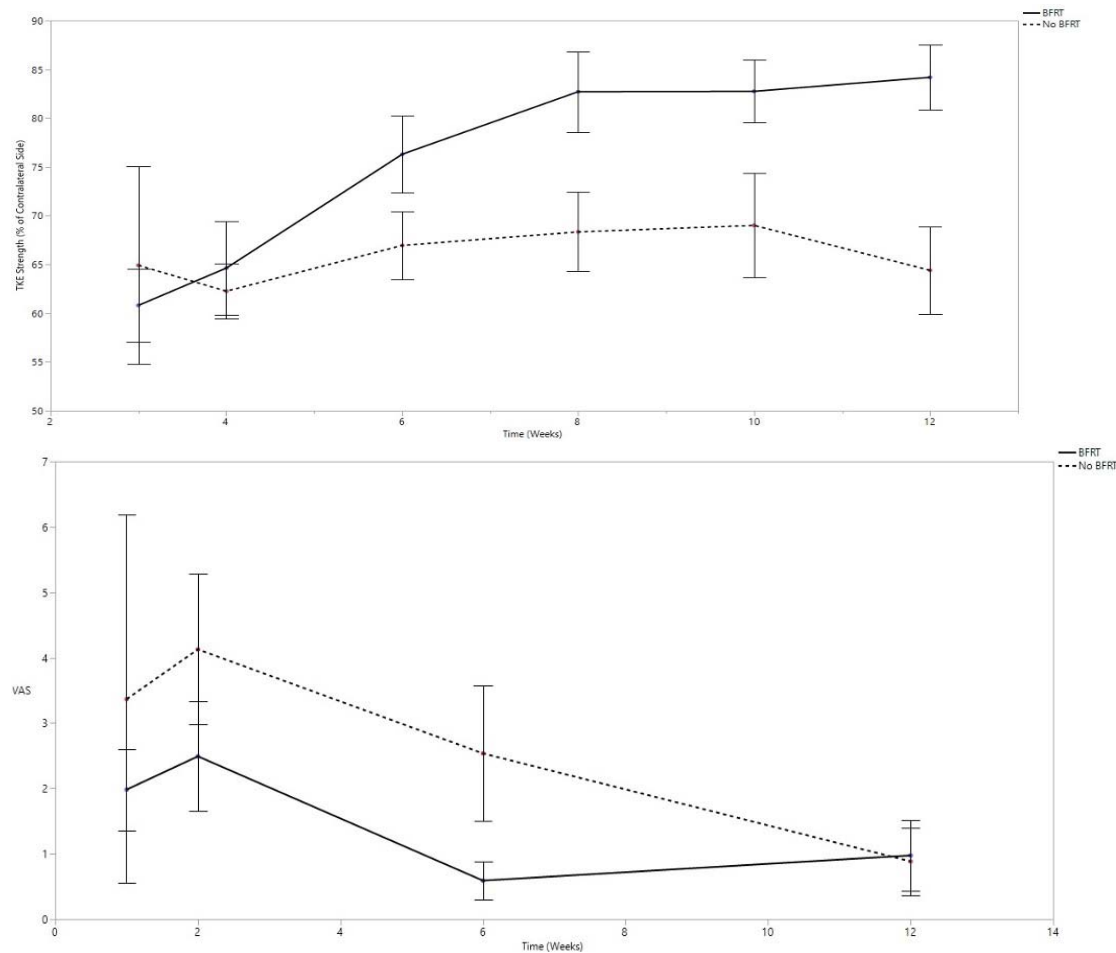


TABLE 2
Change in Outcome Measures From the
Initial Week to the Final Week

Variable	Time of Initial Measurement (Week)	Time of Final Measurement (Week)	Δ Control	Δ BFRT	P-Value
Knee Flexion (degrees)	2	12	56.8	61.0	0.676
Knee Extension (degrees)	2	12	4.4	5.9	0.546
Thigh Circumference (% contralateral side)	2	12	0.4	-1.7	0.606
TKE Strength (% contralateral side)	3	12	8.3	24.4	0.044
VAS Score	0	12	0.6	1.4	0.520
IKDC Score	0	12	6.9	4.6	0.859

None